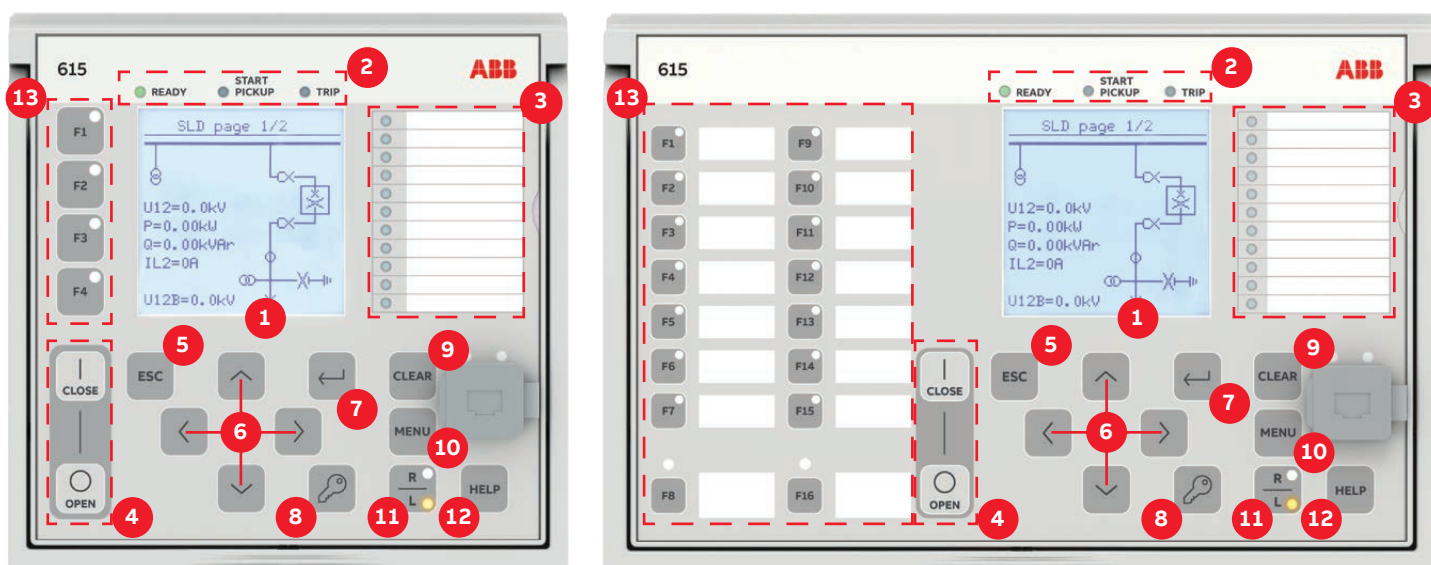


Quick Start Guide

Relion® REX615



1	Display	Default view can be selected from single line diagram (SLD), measurements and main menu
2	Self-supervision and protection indicator LEDs	Ready-LED steady: OK, Ready-LED flashing: Internal Relay Fault (IRF), Start/Pickup-LED steady: protection started, Start/Pickup-LED flashing: protection function blocked, Trip-LED: protection operated
3	Programmable LEDs	Can be programmed for alarming and indication with latching and/or flashing features
4	Control Circuit Breaker (CB)	Press open/close and confirm by pressing enter. If there is more than one controllable object select the object first with navigation buttons. Note: control has to be in Local mode.
5	Escape / Cancel	Used for canceling actions and leaving setting mode without saving the values. Returns back to menu.
6	Navigation	Left = go back, Right = go further, Up = scroll up, Down = scroll down. Up/down can also be used when selecting controllable objects like breakers and switches in single line diagram.
7	Enter / Select	Entering to parameter setting mode and confirming new values
8	Authorization	If authorization is used you can log in or log out using this button
9	Clear	Clearing events and indications, see next page for further details
10	Menu	Switch views in between the main menu, single line diagram and measurements
11	Local / Remote	Changes the control between Local/Remote
12	Help	View help messages
13	Function buttons	Freely configurable

Using the local HMI

Accessing main menu and changing parameters

Press  to navigate between main menu, measurements and single-line diagram.

Press  and select which setting value you want to change and press  to change it with  or  keys. If there is “#” mark at the same line with parameter value, you have to first select which setting group parameter you want to change. If there is no “#” mark then you can change the value directly by pressing  and then with  or  or  or  keys.

You can change all parameters in the same way. The most common function block names are described at the end of this document.

Storing settings

After making changes to parameters they have to be stored to get them into use. Store the settings by going back to main menu by pressing  or using the  key. When IED asks confirmation to commit changes, answer “Yes”. Some changes require the IED to reboot before the changes can be taken into use. Reboot the IED by going to **Menu → Configuration → General → Software reset** or switching the auxiliary power off and back on.

Changing the overcurrent start value

Menu → Settings → Settings → select setting group, default 1 and press  → **Current Protection → PHLPTOC1 → Start value**

Changing function block naming from IEC 61850 names to IEC 60617 or ANSI

Menu → Configuration → HMI → FB Naming convention

Checking binary input value

Menu → Monitoring → I/O Status → Binary input values → select correct BIO card

Checking fault records and the currents and voltages of the latest faults

Menu → Monitoring → Recorded data → Fault record

Changing the display contrast

Hold  and press  or  to change the display contrast. If you want to store the contrast go first in the menu and then go back to the default view, so that A (Administrator) appears to the right upper corner. If authentication is enabled you need to login and then change the contrast.

Clearing events and indications

There are two ways to do this:

- Go to clear menu by pressing  or selecting **Clear** from the main menu. Then select what you want to clear and press  → press  (Clear text will appear and Cancel will disappear) and then press .
- Press  hold for three seconds to clear indications and immediately after that three seconds again to clear LEDs.

Checking IED order code, serial number, HW revision and software version

Menu → Information → Product Identifiers

Display header area

The icon area at the upper right corner of the display shows the current action or user level. These are described below:
S = Parameters are being stored, **!** = Warning and/or indication
V = Viewer, **O** = Operator, **E** = Engineer, **A** = Administrator

I/O status

Menu → Monitoring → I/O status

From here you can check the input and output data of a function block. More information about function block operation can be found under **Monitored data**. Physical input/output statuses, communication monitoring etc are also available.

Changing the language

Menu → Language

Changing the default view

Menu → Configuration → HMI → Default View

Reading the Internal Relay Fault (IRF) code

Menu → Monitoring → IED Status → Self-supervision

Restoring factory settings

Warning! All parameter settings will be overwritten with the default values. During normal use, a sudden change of the settings can cause a protection function to trip.

Menu → Configuration → General → Factory Setting

Using the local HMI

Using the function buttons

REX615 standard includes 4 function buttons and REX615 wide includes 16 function buttons. These buttons can be freely configured by PCM600 to control the relay's internal features for example changing setting group, trigger disturbance recordings and changing operation modes for functions or to control relay's external equipment, for example opening or closing the equipment, via relay's binary outputs.

For REX615 standard press once - LCD shows configured button descriptions, press 2nd - The configured action will be taken.

For REX615 wide check the function button action from the label, before pushing the button. The action will take effect immediately when the button is pressed.

If function buttons are set to follow L/R restriction, the control must be set to Local for button to execute.

Customer documentation is available in the product pages that can be accessed through **abb.com/releon**.



Function block naming

The most common function blocks are listed below, see the full list from the operators manual.

Available function blocks varies depending on the selected application packages and relay configuration. The last digit indicates the instance number of the block, only the first instance is listed below.

IEC 60617	ANSI	IEC 61850	Application package
3Id/I>	87L	LNPLDF	APP6
BST	BST	BSTGAPC	APP6
CVPSOF	SOTF	CVPSOF	APP1
3I>	51P-1	PHLPTOC	APP1
3I>>	51P-2	PHHPTOC	APP1
3I>>>	50P/51P	PHIPTOC	APP1
3I>	51LT	PHLTPTOC	APP1
3I>>>>	50P	PHIPIOC	APP1
3I_3>	51P_3-1	PH3LPTOC	APP1
3I_3>>	51P_3-2	PH3HPTOC	APP1
3I_3>>>	50P_3/51P_3	PH3IPTOC	APP1
3I> ->	67P/51P-1	DPHLPDOC	APP3, APP6, APP8, ADD1, APP10
3I>> ->	67P/51P-2	DPHHPDOC	APP3, APP6, APP8, ADD1, APP10
3I_3> ->	67P_3/51P_3-1	DPH3LPDOC	APP3, APP6, APP8, ADD1, APP10
3I_3>> ->	67P_3/51P_3-2	DPH3HPDOC	APP3, APP6, APP8, ADD1, APP10
Io>	51G/51N-1	EFLPTOC	APP1
Io>>	51G/51N-2	EFHPTOC	APP1
Io>>>	50G/N, 51G/N	EFIPTOC	APP1
Io>>>>	50G/50N	EFIPIOC	APP1
Io> ->	67G/N-1 51G/N-1	DEFLPDEF	APP1
Io>> ->	67G/N-1 51G/N-2	DEFHPDEF	APP1
I1 ->	67P-TC	DPSRDIR	APP3, APP8, ADD1, APP10, APP12
I2 ->, Io ->	67N-TC	DNZSRDIR	APP2, APP3
Yo> ->	21NY	EFPADM	APP2
Io> -> Y	67NYH	MFADPSDE	APP2
Po> ->	32N	WPWDE	APP2
Io> -> IEF	67NTEF/NIEF	INTRPTEF	APP2
Io>HA	51NH	HAEFPTOC	APP2
IF>/UT>	46SNQ/59N	IFPTOC	APP2
I2>	46	NSPTOC	APP1
I2> ->	67Q	DNSPDOC	APP3, APP6, APP10
I2/I1>	46PD	PDNSPTOC	APP1
Uo>	59G/59N	ROVPTOV	APP1, APP5
3U<	27	PHPTUV	APP5
U_A<	27_A	PHAPTUV	APP5
3Urms>	59.S1	PHVPTOV	APP5, APP8
3U>	59	PHPTOV	APP5
U_A>	59_A	PHAPTUV	APP5
U1>	59PS	PSPTOV	APP5
U1<	27PS	PSPTUV	APP5
U2>	59NS	NSPTOV	APP5
f>/f<, df/dt	81	FRPFRQ	APP5

IEC 60617	ANSI	IEC 61850	Application package
3I(U)>	51V	PHPVOC	APP3, APP8, ADD1, APP10
U/f>	24	OEVPVPH	ADD1, APP10
3Ith>F	49F	T1PTTR	APP1
3Ith>T/G/C	49T/G/C	T2PTTR	APP1
3I> 3I<	51, 37, 86C	COLPTOC	APP7
dI>C	60N	CUBPTOC	APP7
3dI>C	60P	HCUBPTOC	APP7
TD>	55ITHD	SRCPTOC	APP7
CNU>	59NU	CNUPTOV	APP7
3dU>	87V	CHPTDV	APP7
U<RT	27RT	LVRTPTUV	APP8
VS	78VS	VVSPPAM	APP8
Q> ->, 3U<	32Q, 27	DQPTUV	APP8
P>/Q>	32R/32O	DOPDPDR	APP3, APP8, ADD1, ADD2, APP10
P<	32U	DUPDPDR	APP3, ADD1, ADD2, APP10
Z<G	21G	UZPDIS	ADD1, APP10
Z2 ->	Z2Q	DNZPDIS	APP3, APP6, APP10
X<	40	UEXPDIS	ADD1
dUo>/Uo3H	64TN	H3EFPSEF	ADD1
Io>R	64R	MREFPTOC	ADD1
3Ith>R	49R	RPTTR	ADD1, ADD2
3dIH>M	87HIM	MHZPDIF	ADD1, ADD2
OOS	78PS	OOSRPSB	ADD1
I2>M	46M	MNSPTOC	APP9
3I<	37	PHPTUC	APP1
3I<	37	LOFLPTUC	APP9
Ist>	50TDJAM	JAMPTOC	APP9
Is2t n<	49, 66, 48, 50TDLR	STTPMSU	APP9
n<	66	MSCPMRI	APP9
I2>>	46R	PREVPTOC	APP9
3Ith>M	49M	MPTTR	APP9
3dI>M/G	87M/87G	MPDIF	ADD1, ADD2
PF<	55U	MPUPF	APP3, APP8, ADD1, APP10
3dI>T	87T	TR2PTDF	APP10
dIoLo>	87NLI	LREFPNDF	APP1
dIoHi>	87NHI	HREFPDIF	APP1
dHi_A>	87_A	HIAPDIF	ADD1, ADD2, APP10, APP11
dHi_B>	87_B	HIBPDIF	ADD1, ADD2, APP10, APP11
dHi_C>	87_C	HICPDIF	ADD1, ADD2, APP10, APP11

IEC 60617	ANSI	IEC 61850	Application package
3I>/Io>BF	50BF	CCBRBRF	APP1
3I2f>	68HB	INRPHAR	APP1
Io2f	68HBG/N	RINRPHAR	APP1
Master Trip	94/86	TRPPTRC	Base
ARC	AFD	ARCSARC	Base
HIF	HIZ	PHIZ	APP2
FLOC	FLOC	SCEFRFLO	APP4
UFLS/R	81LSH	LSHDPFRQ	APP5
MAP	MAP	MAPGAPC	Base
U<, I>	27/50	GAEPVOC	ADD1
LB	21LB	LBRDOB	APP3, APP6
CFD	CFD	RCFD	APP2, APP3
dIo>	87N	CEPNDF	APP1
CLP	62CLD	CLPGAPC	APP1
I>>>CDD	50DD	CDDPTOC	APP1